AEHR - Aehr Test Systems | Deep Dive

Full investment research | Model assumptions and source evidence

Earnings/model calibration: July 14, 2026 · **Engineering, orders and competitive model review:** October 8, 2026
Framework: v8.0; explicit package-level pricing-risk branch
Status: NOT HELD (watchlist, 🔒 CHOKEPOINT) | **Base 93% (56/60)**
Workbook price: $72.60 as of July 13, 2026 | **Diluted shares:** approximately 31.9M
Active Q4 2027 ceiling: $159.93. Q1 2028 rolling forward-four-quarter ceiling: **$185.81**. Portfolio Excel Base and analytical Base are **93/100**.

Supercycle

DB1 Photonics - wafer-level burn-in for silicon photonics and InP devices; AI-processor package-level burn-in through Sonoma; DB4 Power through SiC and GaN.

Core Thesis: Earlier Reliability Screening

Aehr is not simply test equipment. Its economic role is reliability screening for advanced semiconductors operating at increasing power density, thermal stress and system-level cost. As optical I/O and high-speed interconnect production enter an S-shaped adoption curve, test and burn-in capacity must be installed before production devices can ship.

The FOX platform's wafer-level burn-in position remains the central chokepoint thesis. Sonoma adds a competitive but rapidly scaling package-level AI-processor vector. The optical ramp changes the timing of revenue conversion: the model now pulls revenue forward through calendar 2027 rather than relying on unusually high early margins. The framework prices credible breadcrumbs before full production confirmation, then increases or decreases their weight as evidence changes.

FY2026 Q4 Earnings and FY2027 Guidance Anchor

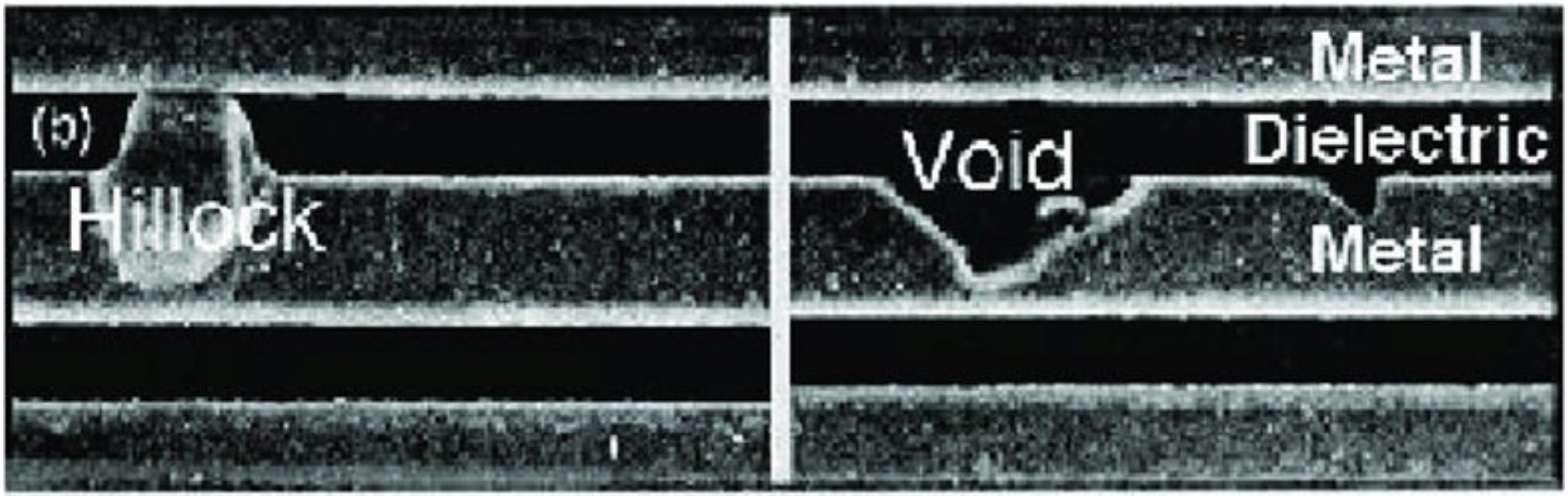
Source: Aehr FY2026 Q4 and full-year results, July 14, 2026

Metric	Reported / guided	Framework interpretation
FY2026 Q4 revenue	$18.8M	Revenue conversion has begun after the bookings inflection.
FY2026 Q4 bookings	$60.7M	Record quarter; approximately 3.2× reported revenue.
Effective backlog	$100.6M	Covers 67-77% of management's FY2027 revenue range.
FY2027 revenue guidance	$130-150M	Management-backed case for current backlog and existing production programs.
FY2027 non-GAAP net income	18-22% of revenue	Validates operating leverage but is not directly interchangeable with operating margin.
Cash, equivalents and restricted cash	$116.5M	Supports manufacturing scale, but the model now uses approximately 31.9M diluted shares.

Management stated that the FY2027 outlook reflects current effective backlog and expected customer demand from existing production programs. Additional orders from current and prospective customers remain outside the fully contracted portion of the guide. The ceiling model therefore treats management's range as the high-confidence core and adds probability-weighted revenue for forecasts, successful benchmarks, second-device work and early memory engagement.

Engineering evidence update - October 6, 2026

Wafer-level screening can avoid committing expensive package components to a processor with an early-life defect. It is not a guarantee that a design ignoring electromigration constraints will survive its mission profile. The March 5 article suggests that designers could relax foundry rules and use a 48-hour screen to reject weak parts. That exact use case is a hypothesis, not a disclosed customer production protocol. Long-duration qualification testing, production screening and wear-out prediction address different populations and time scales. NXP qualification guidance.



Electromigration hillock and void illustration reproduced in IA39, March 5, 2026. This generic image is not an Aehr customer failure analysis or proof of a 48-hour remedy.

The useful economic argument is earlier defect discovery: avoid wasting HBM, interposers, substrates and assembly on unsuitable logic. Screening adds contactor, tool, electricity, cooling, software and service costs, and may reject otherwise usable die. A customer must validate the total yield and reliability trade-off. Assign a 20% research probability to the article's specific relaxed-design-rule use case becoming commercially material beyond ordinary screening; no named program or measurable revenue scope supports a separate increment. Ordinary AI screening is already in the approved ramp.

Later production orders strengthen the existing ramp

Aehr announced a **$22M follow-on order on August 12** from its existing lead AI wafer-level customer, with systems expected to ship over the following six months. It covers FOX-XP systems, WaferPak contactors and aligners. This is not the newly benchmarked second customer's production award. Aehr order release.

An **August 4 optical follow-on** calls for one FOX-XP with nine independently powered WaferPak blades, with shipment expected in H1 2027. The stated power capability is 3,500W per blade. It is not a disclosed annual wafer throughput or system selling price. Aehr optical order.

Probability, overlap and cost bridge: give the $22M order a 90% shipment/recognition timing weight for its announced window, with an 85-100% diagnostic range. The central $19.8M supported revenue attribution is mapped to the lead-customer expansion already anticipated in the model. This mapping is an analytical overlap judgment, **not a disclosed order-by-order reconciliation**. At the retained 18-29% quarterly operating-margin range across the expected shipping window, the weighted attribution corresponds to $3.564-5.742M of OP and $14.058-16.236M of combined operating costs inside the current forecast. Those endpoints reflect timing/mix attribution, not a newly adopted margin. There is no extra $22M or $19.8M added to total revenue. An 85% research probability is assigned to the optical system shipping in H1 2027; its undisclosed invoice value remains a precise gap.

The FY2027 model window, calendar Q3 2026 through Q2 2027, is now **$179.874750M** after the bounded package-level pricing adjustment below, versus management's $130-150M guidance. The remaining **$29.87-49.87M forecast gap** must be earned through delivery and acceptance. Bookings after July cannot simply be added to July backlog without reconciling shipments and acceptances.

Calendar-2027/2028 active revenue is **$265.457750M/$498.500000M**. Reference volume and margin assumptions remain visible; active pricing and OP include the risk once. The model retains an OP-margin proxy without inventing separate gross-profit and operating-expense forecasts. The 30× F4Q OP method, 31.9M shares and Base 93 are unchanged.

Package-level orders and competition - October 8, 2026

On October 7, Aehr announced approximately **$6M** of follow-on Sonoma systems, burn-in modules, sockets and software orders through a Taiwan OSAT for its existing lead hyperscale customer's second AI processor. Delivery is expected over six months. Its initial second-processor win was announced in February; these orders support the same program's calendar-2027 ramp. This is an order update, not another earnings release or a new customer.

Assign **90% probability** to timely shipment and recognition, with an 85-100% diagnostic band. The central **$5.4M** revenue attribution overlaps the already modeled second-device ramp **100%**. At 24-33% reference OP margins in the shipping window, it supports **$1.296-1.782M OP** and **$3.618-4.104M costs inside the model**. Those are timing/mix attributions, not additional revenue or a newly adopted product margin. Quarterly acceptance and the systems/consumables mix remain unknown. Do not subtract the competitive scenario from this signed order again.

Aehr order release, October 7 and initial second-processor award, February 11.



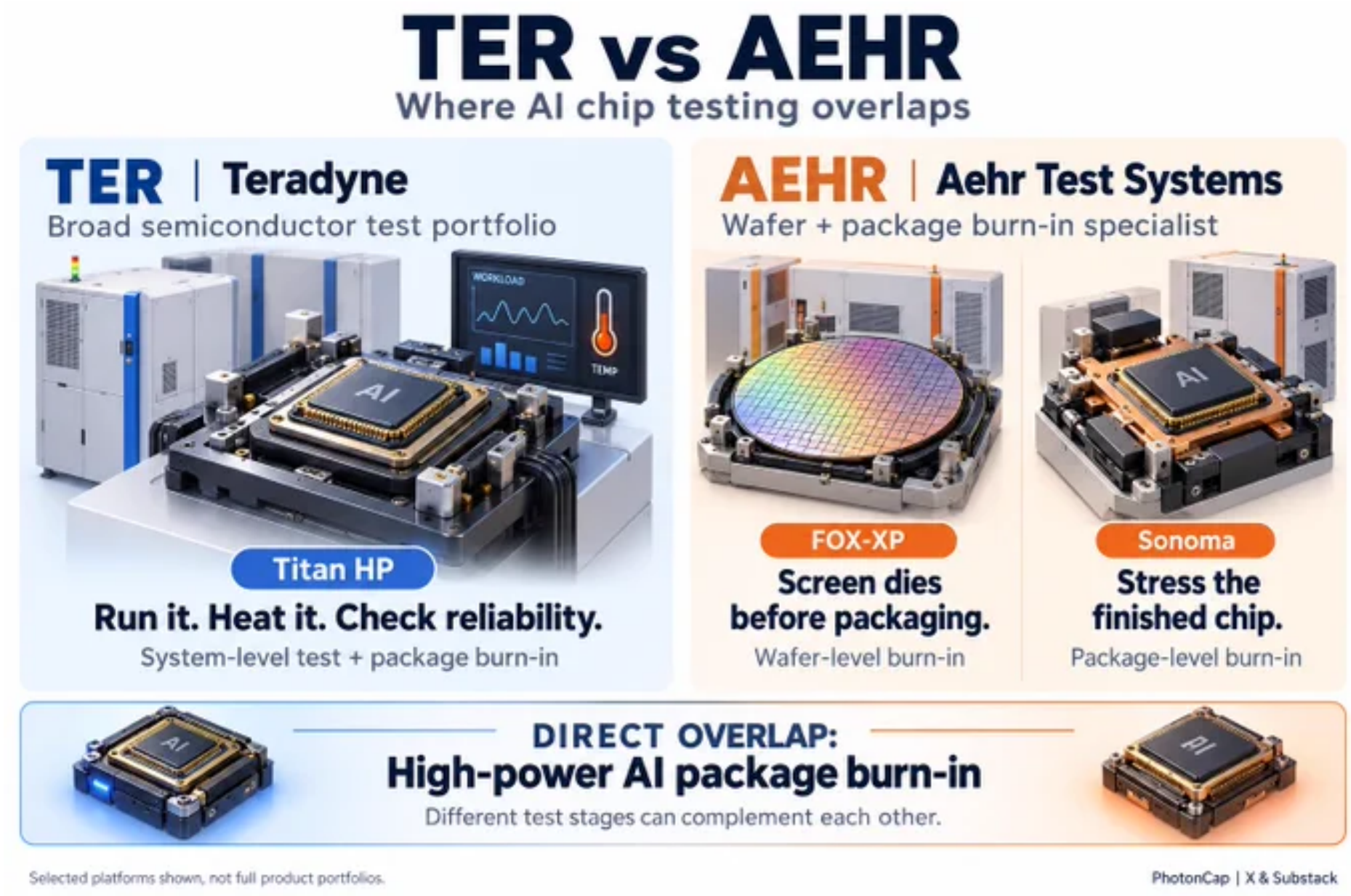
Aehr issuer release header, October 7, 2026. The second processor is a second program for the same lead customer; no quarterly revenue guidance is supplied.

Where Titan HP competes

Teradyne's October 6 production-deployment release combines package burn-in and system-level test on Titan HP, with independent device thermal control and kilowatt-class power. It can reduce handling steps and floor space. Those capabilities strengthen competition for **Sonoma package-level programs**. They do not establish a replacement for **FOX wafer-level burn-in before packaging**. No named Aehr customer loss, comparative yield/throughput benchmark or realized competitor price was disclosed.

Sonoma also has independent thermal zones and individual socket control for complex high-power devices. The commercial comparison is accepted throughput, reliability and total cost for a particular workload. Wafer screening and package-level testing can be separate, complementary insertions.

Teradyne issuer release, October 6; Aehr package burn-in capabilities.



PhotonCap stage diagram circulated October 7, 2026. This is not an issuer benchmark: Titan overlaps Sonoma after packaging; FOX screens at wafer level. No measured customer win/loss, price or throughput comparison.

The active pricing-risk branch

Use **30% probability** of a **5% net price concession on 20% of future revenue**. The 20% is a **LOW-confidence analytical bound for contestable Sonoma/new orders**, not a disclosed Sonoma segment share. Generic competition was already recognized in the thesis and reference margins; this branch represents additional pressure beyond that reference from the production-deployment evidence. It is the one explicit incremental pricing deduction, with no second blanket risk or volume cut.

The phase is zero in 2026, then 25%, 50%, 75% and 100% across 2027, remaining 100% thereafter as a longer-range assumption. New/repriceable orders take time to convert to shipments. FOX wafer-level volume and the booked $6M are outside the intended exposure. The aggregate model cannot assign every customer dollar, making the exposure proxy a material uncertainty.

Weighted revenue loss = reference revenue × 30% × 20% × 5% × phase. Lost price provides little production-cost relief: assume expense savings of **5% of lost revenue**, leaving **95% as lower OP**. No system-unit loss, share issuance or valuation-multiple change is added. Reference operating costs remain except for these savings; effective margins move modestly instead of staying constant on lower price.

Diagnostic bounds are 25-35% probability, 10-30% exposure and 3-7% concession. They are analyst sensitivities, not measured discounts. Reweight with accepted competitive wins, realized new-order pricing, Sonoma mix and comparable reliability/throughput. Aehr follow-on orders, its distinct wafer-level business and a growing burn-in market argue against a company-wide displacement claim.

Quarterly competition bridge

Quarter	Reference revenue $M	Phase	Revenue loss $M	Expense savings $M	OP loss $M
Q3 2026	31.000	0%	0.000000	0.000000	-0.000000
Q4 2026	40.000	0%	0.000000	0.000000	-0.000000
Q1 2027	51.000	25%	0.038250	0.001913	0.036338
Q2 2027	58.000	50%	0.087000	0.004350	0.082650
Q3 2027	72.000	75%	0.162000	0.008100	0.153900
Q4 2027	85.000	100%	0.255000	0.012750	0.242250
Q1 2028	100.000	100%	0.300000	0.015000	0.285000
Q2 2028	115.000	100%	0.345000	0.017250	0.327750
Q3 2028	135.000	100%	0.405000	0.020250	0.384750
Q4 2028	150.000	100%	0.450000	0.022500	0.427500
Q1 2029	150.000	100%	0.450000	0.022500	0.427500
Q2 2029	155.000	100%	0.465000	0.023250	0.441750
Q3 2029	165.000	100%	0.495000	0.024750	0.470250
Q4 2029	180.000	100%	0.540000	0.027000	0.513000
Q1 2030	185.000	100%	0.555000	0.027750	0.527250
Q2 2030	195.000	100%	0.585000	0.029250	0.555750
Q3 2030	205.000	100%	0.615000	0.030750	0.584250
Q4 2030	215.000	100%	0.645000	0.032250	0.612750

Q4 2027-Q3 2028 includes **$1.305000M lower revenue**, **$0.065250M expense savings** and **$1.239750M lower OP**. Active F4Q OP is **$170.060250M**. At 30× and 31.9M diluted shares, the ceiling is **$159.931270**. Base remains 93: package competition was already part of the thesis assessment and now has an explicit earnings cost.

Optical S-Curve Evidence

1. The lead silicon-photonics customer is ramping and has already placed follow-on system orders.
2. A new major silicon-photonics customer described as a global networking leader has forecast additional systems during calendar 2026.

3. The lead AI wafer-level burn-in customer is shifting accelerator burn-in from system level to all wafer level and is significantly expanding capacity.
4. A second top-tier supplier of AI accelerators, CPUs and network processors completed benchmark testing above expectations, expressed interest in pilot production validation and requested evaluation of a second device.
5. Optical test capacity must precede volume device shipments, so Aehr bookings and system installations are leading indicators for the optical production curve.

Breadcrumb-Weighting Method

The model does not wait for a production award to recognize all future revenue. That would systematically under-model a business whose stock can gap before the award becomes visible in reported results.

Probability-weighted revenue = potential revenue × conversion probability × timing probability

Evidence state	Current modeling weight	Interpretation
Effective backlog and current production programs	90-100%	Highest-confidence conversion base.
Additional systems forecast by a new networking customer	70-80%	Customer forecast is meaningful evidence, but timing and final configuration can move.
AI-processor benchmark exceeded expectations plus pilot interest	50-60%	Commercial probability is recognized before a production award.
Second AI device requested for evaluation	25-35%	Earlier-stage option with lower conversion and timing confidence.
NAND benchmark completed and development agreement under discussion	30-40%	Technical progress is real; production timing and customer identity remain unconfirmed.
HBM production revenue	15-25%	Strategic option only until a development or production commitment improves visibility.

These are evidence weights, not permanent risk discounts applied to the entire company. Each stream is reweighted independently as orders, customer forecasts, qualifications, delays or cancellations arrive.

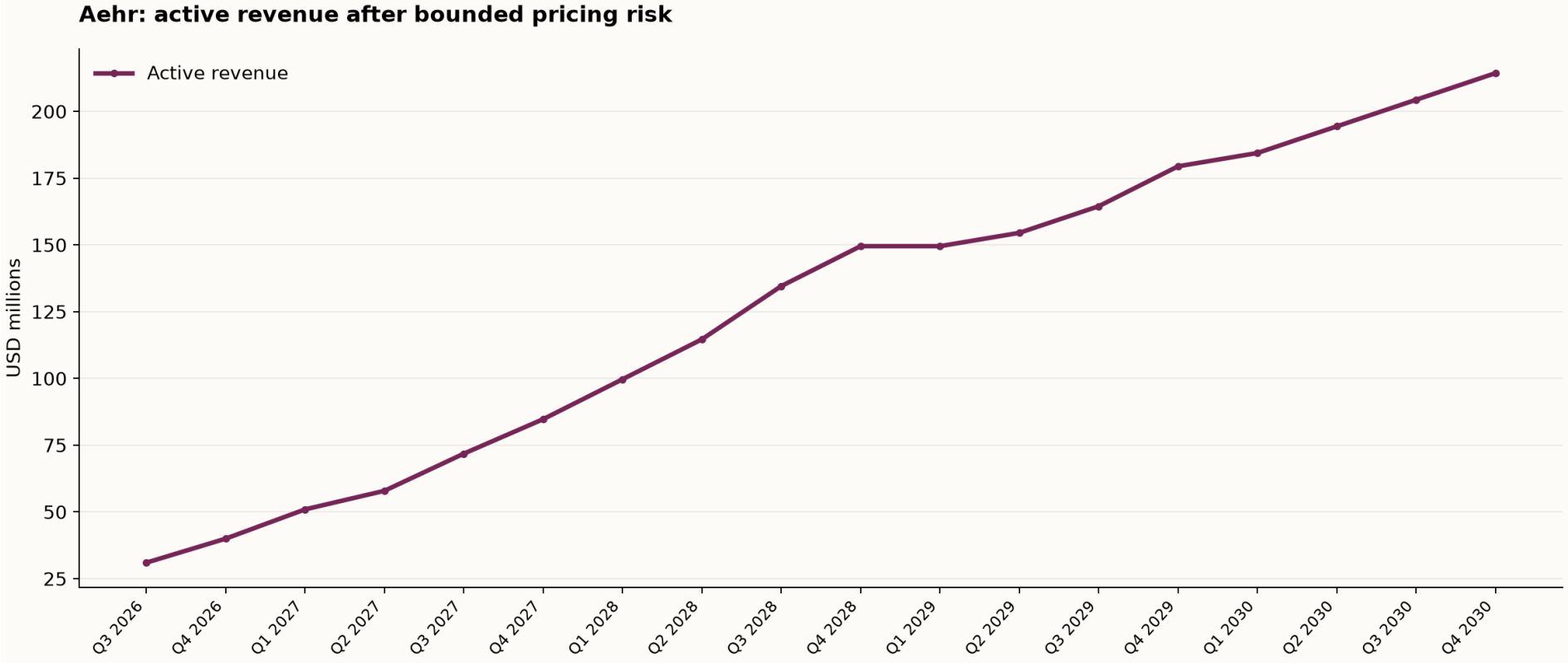
Approved Probability-Weighted Rolling Forward-4Q Ceiling Path

Formula:

Quarterly ceiling = operating profit for the current quarter and next three quarters × 30× ÷ 31.9M diluted shares

Calendar quarter	Revenue $M	OP margin	OP $M	Forward-4Q OP $M	Ceiling / share
Q3 2026	31.000	18.00%	5.580	48.991	$46.07
Q4 2026	40.000	24.00%	9.600	69.177	$65.06
Q1 2027	50.962	28.95%	14.754	91.635	$86.18
Q2 2027	57.913	32.91%	19.057	115.596	$108.71
Q3 2027	71.838	35.87%	25.766	142.211	$133.74
Q4 2027	84.745	37.83%	32.058	170.060	$159.93
Q1 2028	99.700	38.83%	38.715	197.575	$185.81
Q2 2028	114.655	39.83%	45.672	218.433	$205.42
Q3 2028	134.595	39.83%	53.615	234.319	$220.36
Q4 2028	149.550	39.83%	59.572	246.233	$231.57

Financial view 1. Revenue and mix



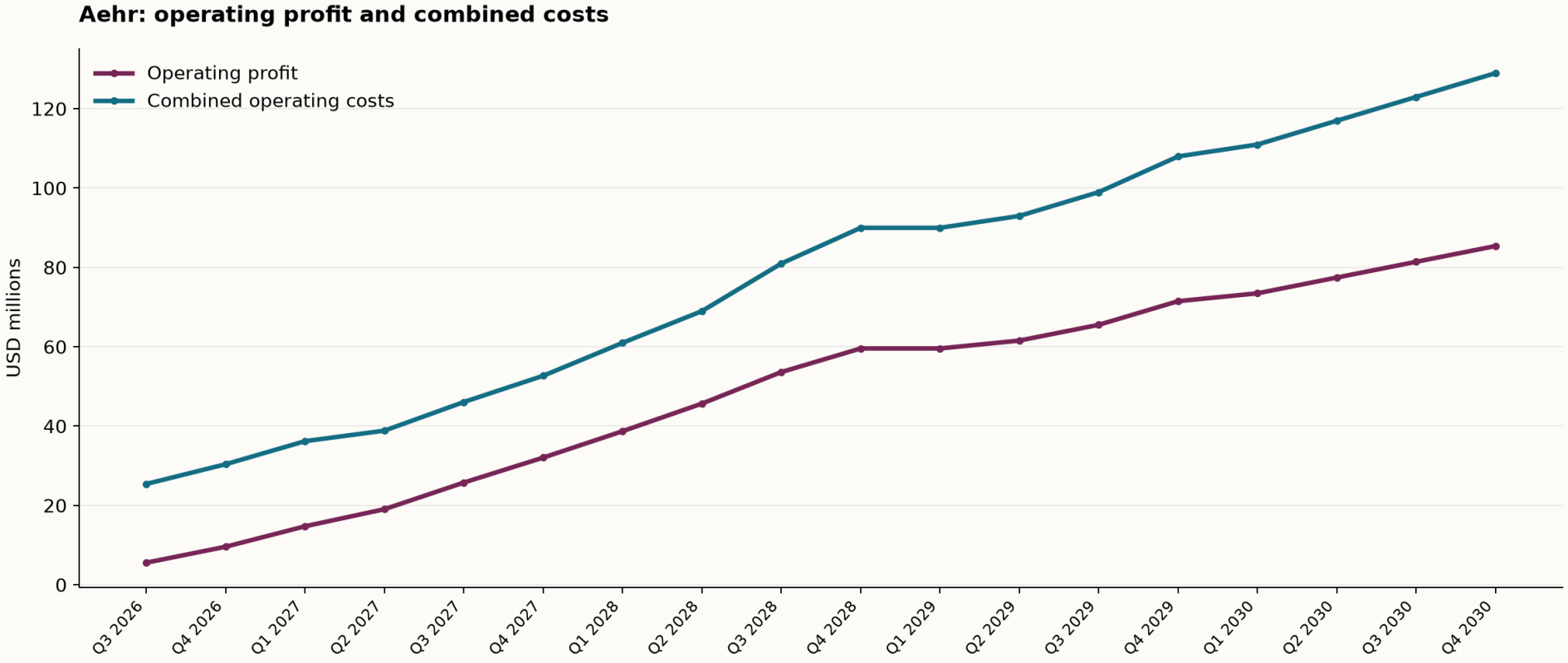
Aehr: active revenue after bounded pricing risk. USD millions. Model as of 8 October 2026.

Future rows are Model / LOW forecast confidence. Product sales are not split in the sanctioned aggregate model; 20% contestable exposure is a risk proxy, not segment disclosure.

Quarter	Reference revenue $M	Active revenue $M
Q3 2026	31.000000	31.000000
Q4 2026	40.000000	40.000000
Q1 2027	51.000000	50.961750
Q2 2027	58.000000	57.913000
Q3 2027	72.000000	71.838000

Quarter	Reference revenue $M	Active revenue $M
Q4 2027	85.000000	84.745000
Q1 2028	100.000000	99.700000
Q2 2028	115.000000	114.655000
Q3 2028	135.000000	134.595000
Q4 2028	150.000000	149.550000
Q1 2029	150.000000	149.550000
Q2 2029	155.000000	154.535000
Q3 2029	165.000000	164.505000
Q4 2029	180.000000	179.460000
Q1 2030	185.000000	184.445000
Q2 2030	195.000000	194.415000
Q3 2030	205.000000	204.385000
Q4 2030	215.000000	214.355000

Financial view 2. Margins, costs and profit



Aehr: operating profit and combined costs. USD millions. Model as of 8 October 2026.

Gross profit and separate opex remain unavailable in the quarterly proxy. Combined operating costs equal revenue less OP and include the explicit expense savings.

Quarter	Revenue $M	GP	Opex	OP $M	Effective OPM	All costs $M
Q3 2026	31.000000	Not split	Not split	5.580000	18.000%	25.420000
Q4 2026	40.000000	Not split	Not split	9.600000	24.000%	30.400000
Q1 2027	50.961750	Not split	Not split	14.753662	28.950%	36.208088
Q2 2027	57.913000	Not split	Not split	19.057350	32.907%	38.855650
Q3 2027	71.838000	Not split	Not split	25.766100	35.867%	46.071900
Q4 2027	84.745000	Not split	Not split	32.057750	37.828%	52.687250

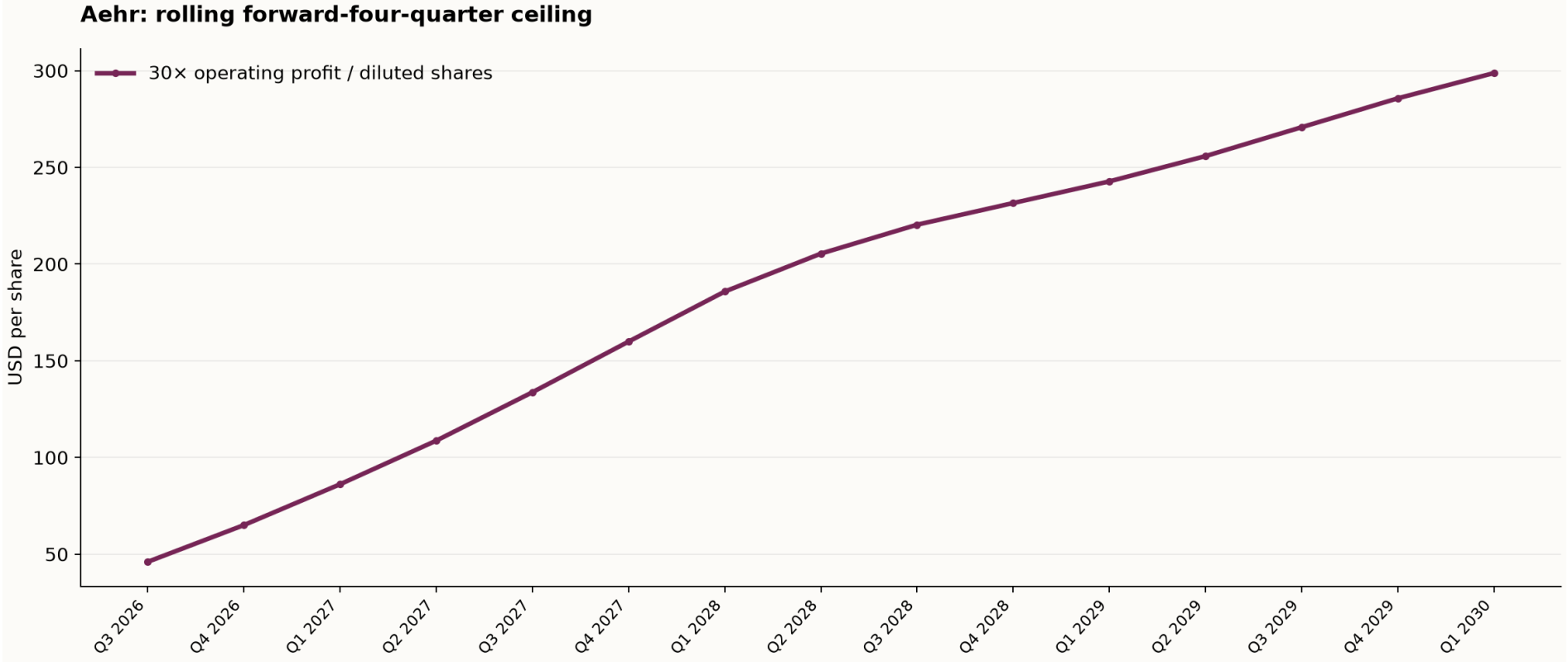
Quarter	Revenue $M	GP	Opex	OP $M	Effective OPM	All costs $M
Q1 2028	99.700000	Not split	Not split	38.715000	38.831%	60.985000
Q2 2028	114.655000	Not split	Not split	45.672250	39.835%	68.982750
Q3 2028	134.595000	Not split	Not split	53.615250	39.835%	80.979750
Q4 2028	149.550000	Not split	Not split	59.572500	39.835%	89.977500
Q1 2029	149.550000	Not split	Not split	59.572500	39.835%	89.977500
Q2 2029	154.535000	Not split	Not split	61.558250	39.835%	92.976750
Q3 2029	164.505000	Not split	Not split	65.529750	39.835%	98.975250
Q4 2029	179.460000	Not split	Not split	71.487000	39.835%	107.973000
Q1 2030	184.445000	Not split	Not split	73.472750	39.835%	110.972250
Q2 2030	194.415000	Not split	Not split	77.444250	39.835%	116.970750
Q3 2030	204.385000	Not split	Not split	81.415750	39.835%	122.969250
Q4 2030	214.355000	Not split	Not split	85.387250	39.835%	128.967750

Annual Revenue and Operating-Profit Envelope

Year	Revenue $M	OP $M	OP margin
2027	265.457750	91.634862	34.520%
2028	498.500000	197.575000	39.634%
2029	648.050000	258.147500	39.835%
2030	797.600000	317.720000	39.835%

The FY2027 calendar Q3 2026-Q2 2027 model is $179.874750M. Original volume and reference margin assumptions remain; realized-price risk lowers active revenue and OP. Mature reference margin remains 40%, while active margin includes competition costs. Later years are low-confidence continuation assumptions.

Financial view 3. Complete active valuation path



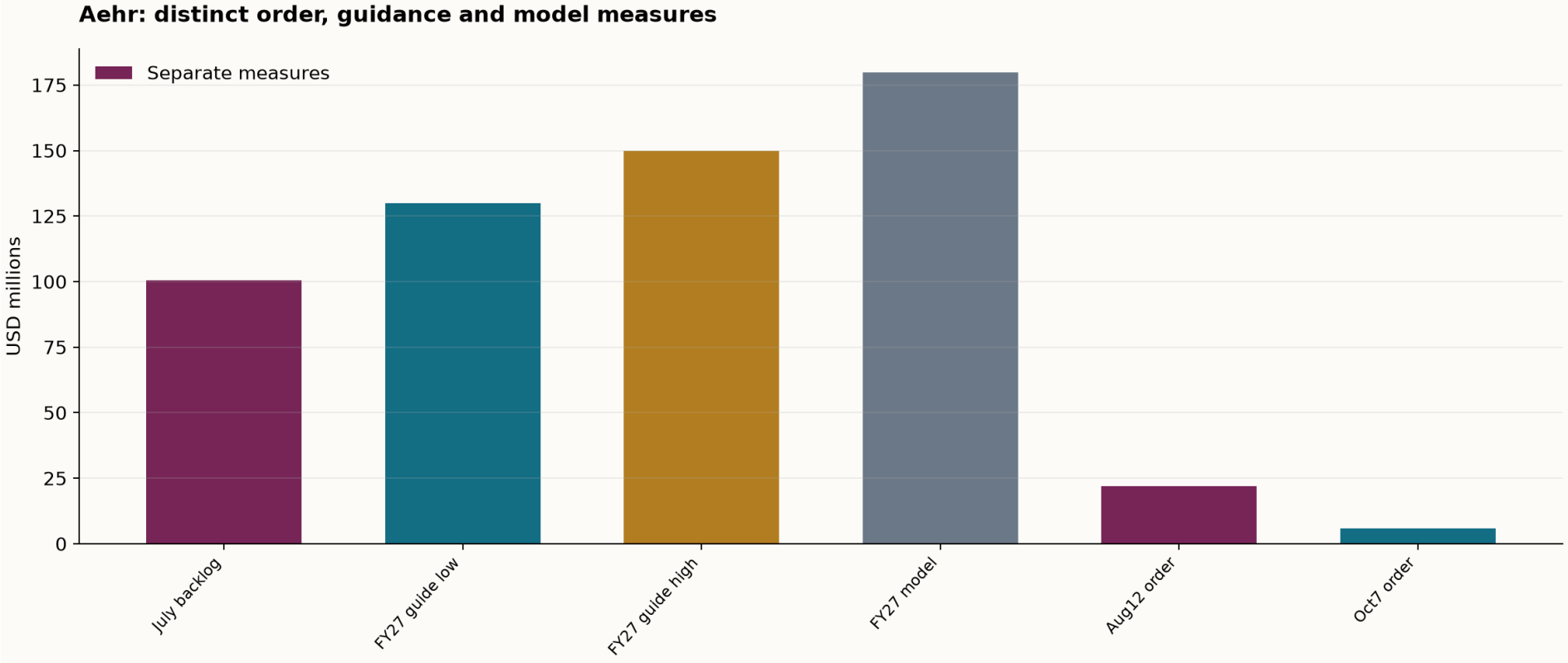
Aehr: rolling forward-four-quarter ceiling. USD per share. Model as of 8 October 2026.

The active horizon begins Q4 2027. The first quarter plus the next three forms each window. Shares, multiple, FX/unit factor and post-weight factor are unchanged.

Quarter	Window end	F4Q OP $M	Multiple	Shares M	Ceiling
Q3 2026	Q2 2027	48.991012	30	31.900000	46.073053
Q4 2026	Q3 2027	69.177112	30	31.900000	65.056846
Q1 2027	Q4 2027	91.634862	30	31.900000	86.176987
Q2 2027	Q1 2028	115.596200	30	31.900000	108.711160
Q3 2027	Q2 2028	142.211100	30	31.900000	133.740846
Q4 2027	Q3 2028	170.060250	30	31.900000	159.931270

Quarter	Window end	F4Q OP $M	Multiple	Shares M	Ceiling
Q1 2028	Q4 2028	197.575000	30	31.900000	185.807210
Q2 2028	Q1 2029	218.432500	30	31.900000	205.422414
Q3 2028	Q2 2029	234.318500	30	31.900000	220.362226
Q4 2028	Q3 2029	246.233000	30	31.900000	231.567085
Q1 2029	Q4 2029	258.147500	30	31.900000	242.771944
Q2 2029	Q1 2030	272.047750	30	31.900000	255.844279
Q3 2029	Q2 2030	287.933750	30	31.900000	270.784091
Q4 2029	Q3 2030	303.819750	30	31.900000	285.723903
Q1 2030	Q4 2030	317.720000	30	31.900000	298.796238

Financial view 4. Orders, guidance and conversion gap



Aehr: distinct order, guidance and model measures. USD millions. Model as of 8 October 2026.

Different dates and scopes are not additive. Later orders support existing programs; reconcile shipments and acceptances before comparing them with July backlog.

Evidence / assumption	Amount USD M	Recognition / qualification
July effective backlog	100.600	Reported July14; multiple periods
FY2027 guidance	130-150	Management July14
FY2027 model	179.874750	Q3 2026-Q2 2027 active forecast
Gap above guidance	29.875-49.875	Remaining model risk
Aug12 lead-AI order	22.000	Next six months; overlaps wafer-level ramp
Aug12 timing attribution	19.800	p90%; already within model

Evidence / assumption	Amount USD M	Recognition / qualification
Oct7 Sonoma order	6.000	Next six months; same customer second program
Oct7 timing attribution	5.400	p90%; 100% overlap; no new revenue

Revenue Vectors

V1 - Silicon-Photonics Wafer-Level Burn-In: Primary

The optical production curve is the primary timing change. Lead-customer follow-on systems, the second networking customer and broad AI interconnect demand justify pulling the prior 2028 revenue opportunity into 2026-2027. The ceiling assumes production conversion, not immediate universal adoption.

V2 - Sonoma and AI-Processor Burn-In: High, Competitive

The lead hyperscale customer is forecasting expansion for a second device with approximately twice the package power of the first. Package-level burn-in remains competitive with Advantest and Teradyne, so V2 does not receive a monopoly assumption. The active model includes the bounded Titan HP pricing risk above. The newly benchmarked top-tier processor supplier is probability-weighted before a completed production award; its weight should rise materially if pilot validation becomes a production program.

V3 - SiC and GaN Power: Medium

Aehr reported approximately $8M of new SiC WaferPak orders in the month preceding the results. This restores the power-semiconductor vector but does not drive the optical S-curve revision.

V4 - NAND and HBM: Early Probability-Weighted Optionality

The completed NAND benchmark and development-agreement discussion justify a small probability-weighted contribution. The customer is not identified in the model, and the framework does not assume the unconfirmed SNDK attribution. HBM remains lower-weight optionality until the evidence improves.

Scoring

Category	Score	July 14 assessment
Core Thesis	20/20	Wafer-level burn-in remains the central reliability chokepoint.
Cycle Stage	9/10	Record bookings and $100.6M effective backlog confirm the conversion phase.
TAM	9/10	Optical, AI-processor and power vectors are expanding; package-level competition caps the score.
Catalysts	5/5	Optical system conversions, second AI-customer pilot and next-generation device ramps.
Macro	10/10	AI interconnect bandwidth and optical production intensity remain structurally favorable.
Alpha	4/5	The bookings-to-revenue lag and test-before-ship timing remain underappreciated.
Risks	-1	Customer concentration and conversion timing remain material.
Base	**56/60 - 93%**	Unchanged. The earnings release validates the revenue ramp but does not remove concentration risk.

Risks and Falsification Tests

1. **Optical qualification timing:** system forecasts fail to convert into orders or installations over the next four quarters.
2. **Backlog conversion:** FY2027 revenue falls below management's $130M low end without an explicit customer-timing explanation.
3. **Margin conversion:** consumables mix and manufacturing leverage fail to move operating margin toward the modeled ramp.
4. **Customer concentration:** the lead AI or optical customer materially reduces its capacity forecast.
5. **Competition:** a competitor commercially ships a directly substitutable high-volume wafer-level burn-in platform.
6. **Package-level competition:** Advantest or Teradyne limits Sonoma share or pricing.

Re-Rate Triggers

1. Production order from the newly benchmarked major AI-processor supplier.
2. Quantified volume forecast or purchase order for its second evaluated device.
3. Additional FOX-XP production orders from either silicon-photonics customer.
4. Evidence that consumables exceed 30% of revenue as the installed base expands.
5. FY2027 revenue tracking above the active $179.87M modeled window without margin deterioration.

A production award from the second major AI supplier would materially raise its conversion weight and could justify moving the ceiling above the current approximately $186 Q1 2028 strategic level. A quantified customer forecast, successful pilot, additional device request or development agreement can also raise the model before a

production order, but by less because timing and conversion remain uncertain.

Current model controls

All quarterly operating figures are model estimates, not reported actuals. Revenue and OP incorporate the selected competition branch once. Valuation remains 30× forward-four-quarter OP divided by 31.9M diluted shares. Analytical and portfolio Excel Base are 93/100. Prior versions are preserved outside this clean current artifact.